

PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL (Create index, Create view, Analyze Table)



Oleh:

ZAKARIA

103122400042

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

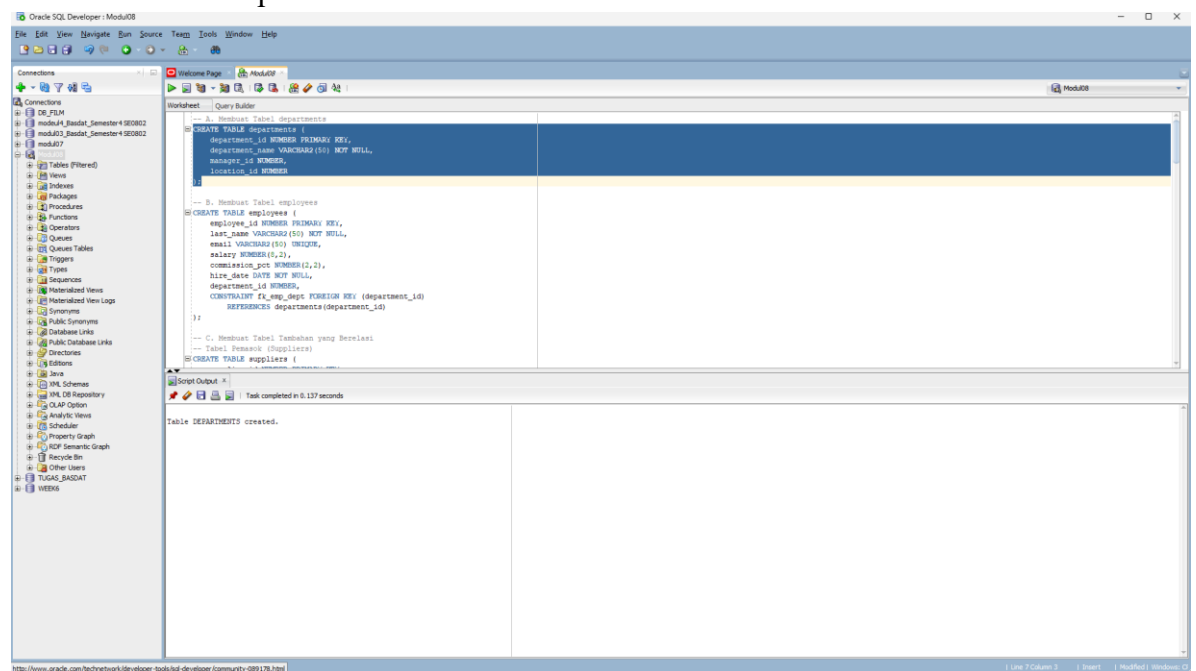
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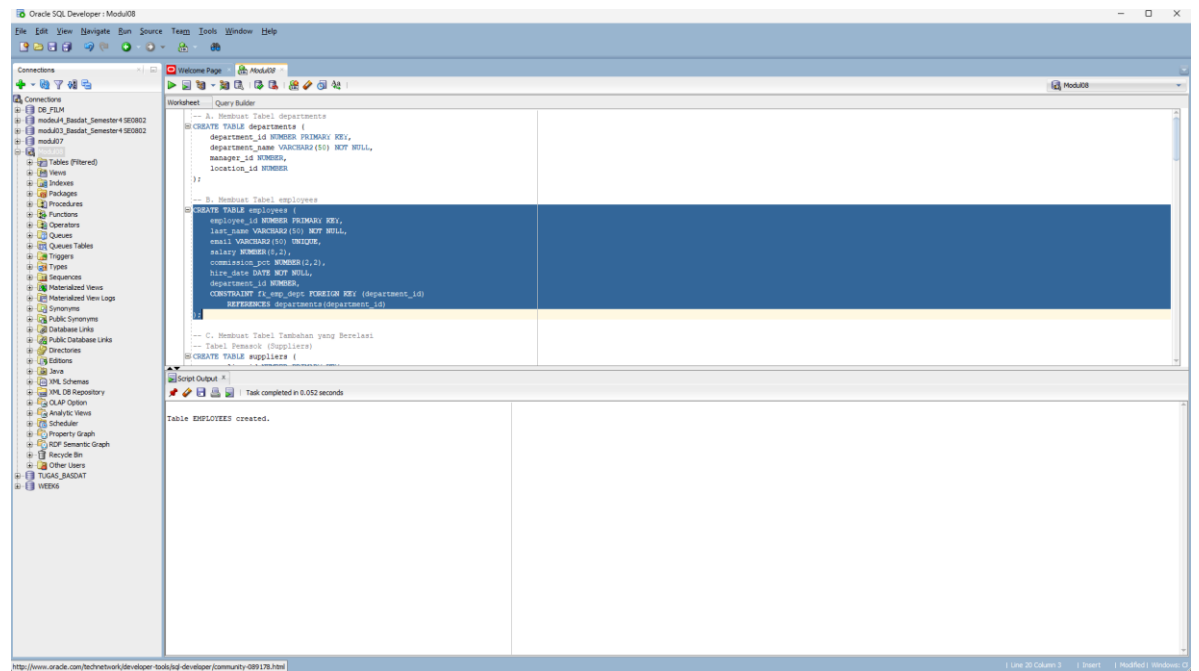
1. Buatlah desain database untuk suatu perusahaan:
 - a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
 - b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
 - c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

Jawaban:

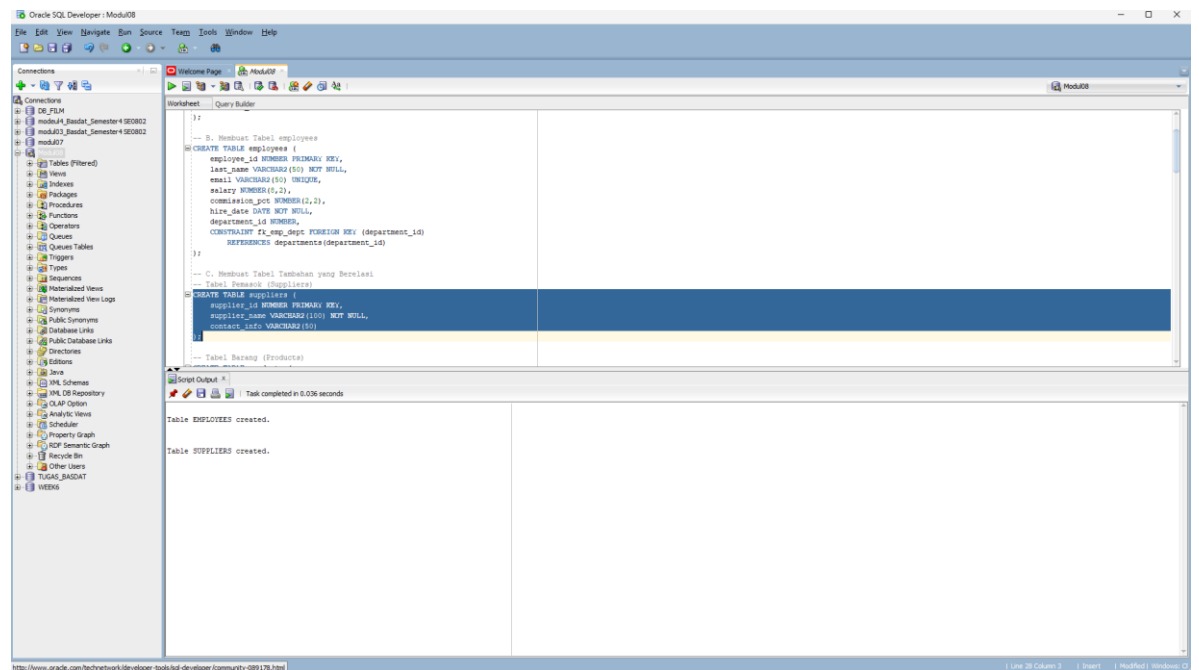
a. Membuat Tabel Departement

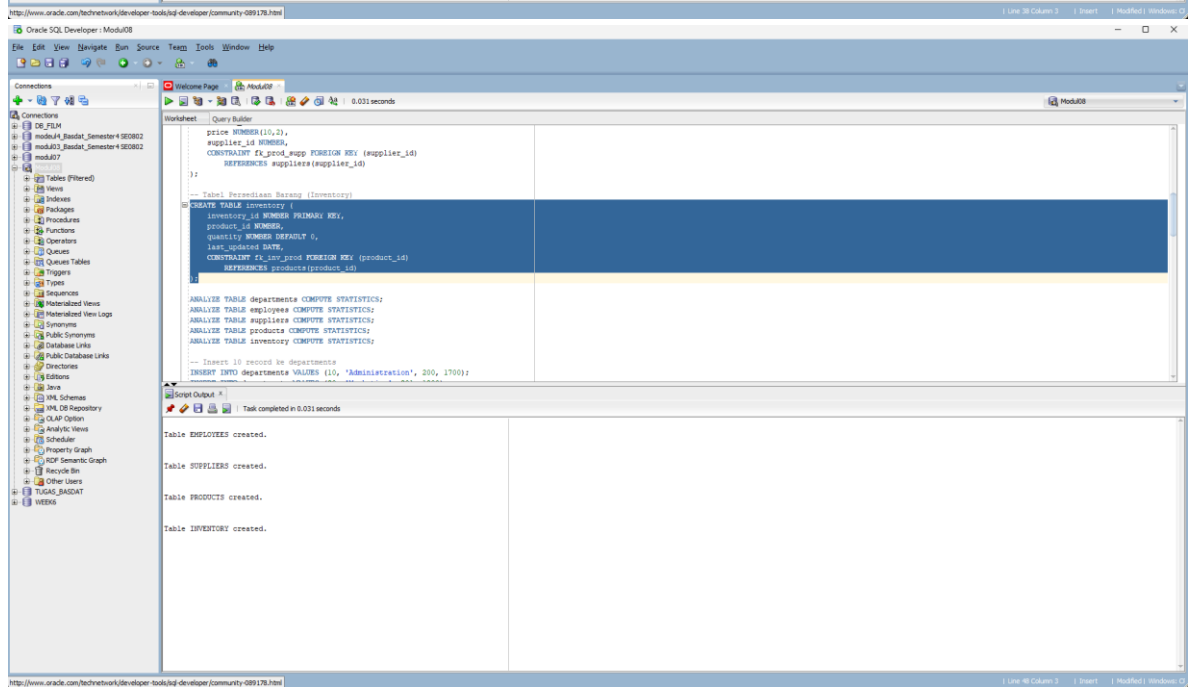
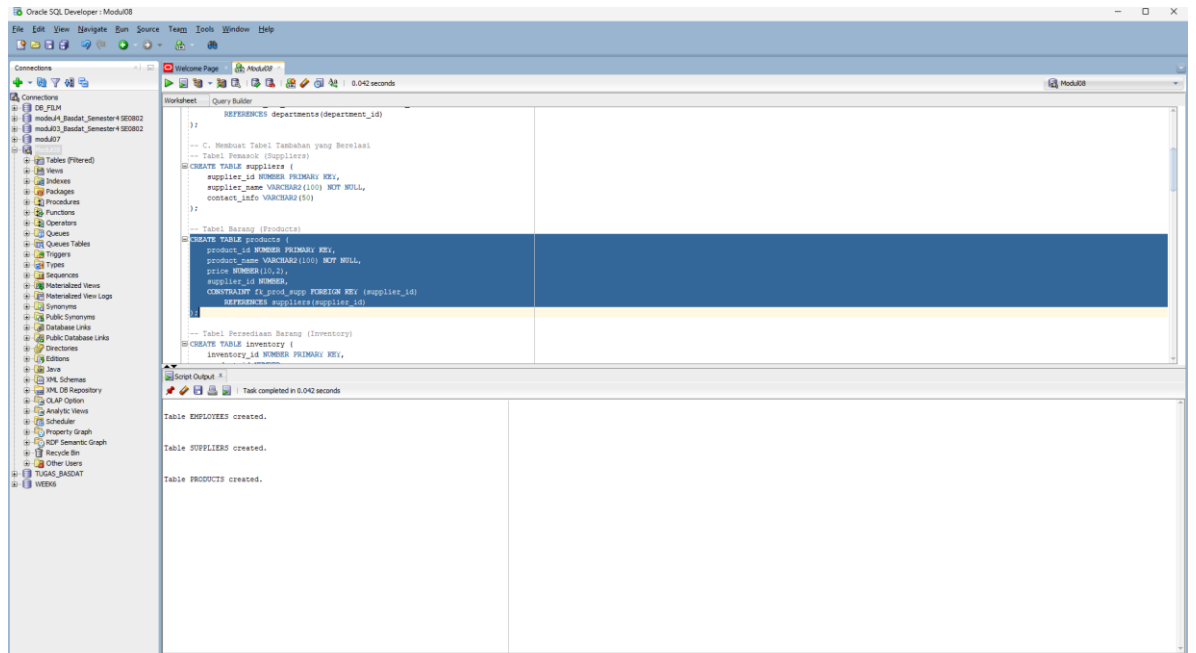


b. Membuat Tabel Employee



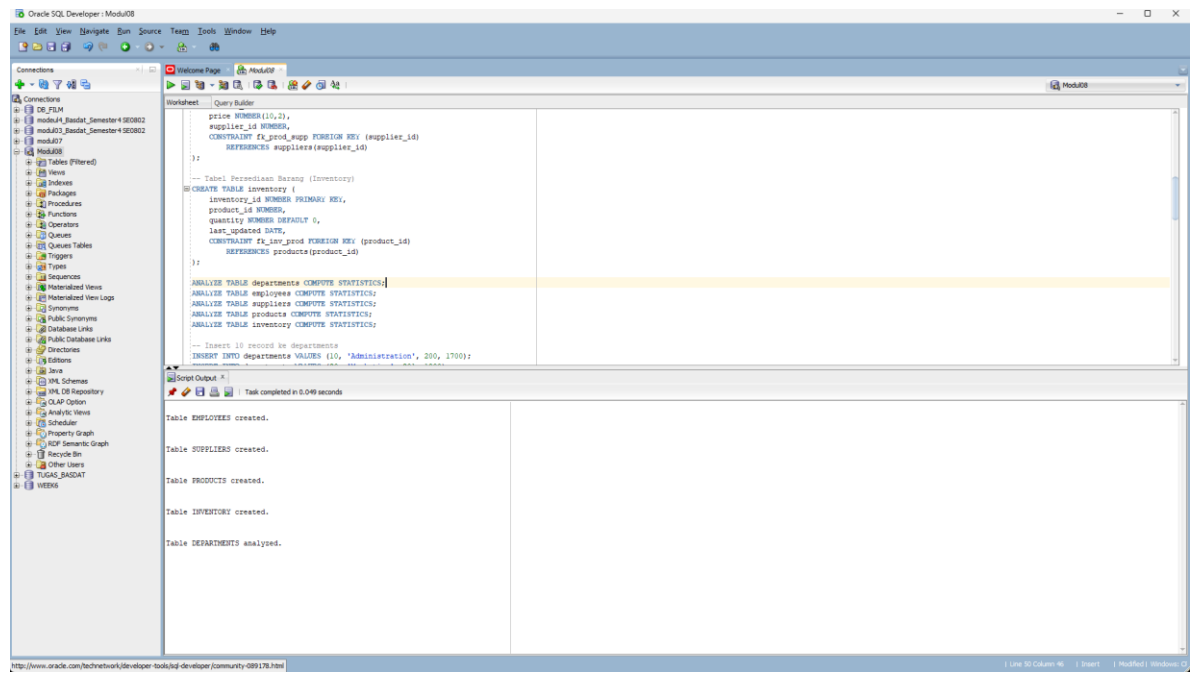
c. Membuat Tabel Tambahan

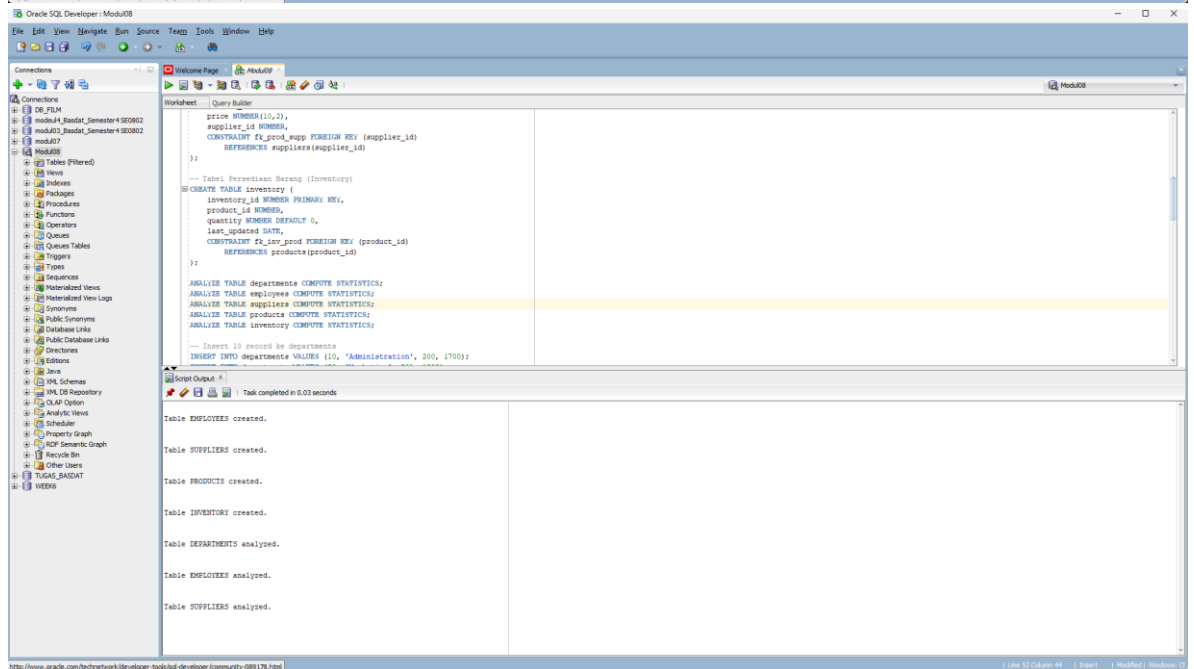
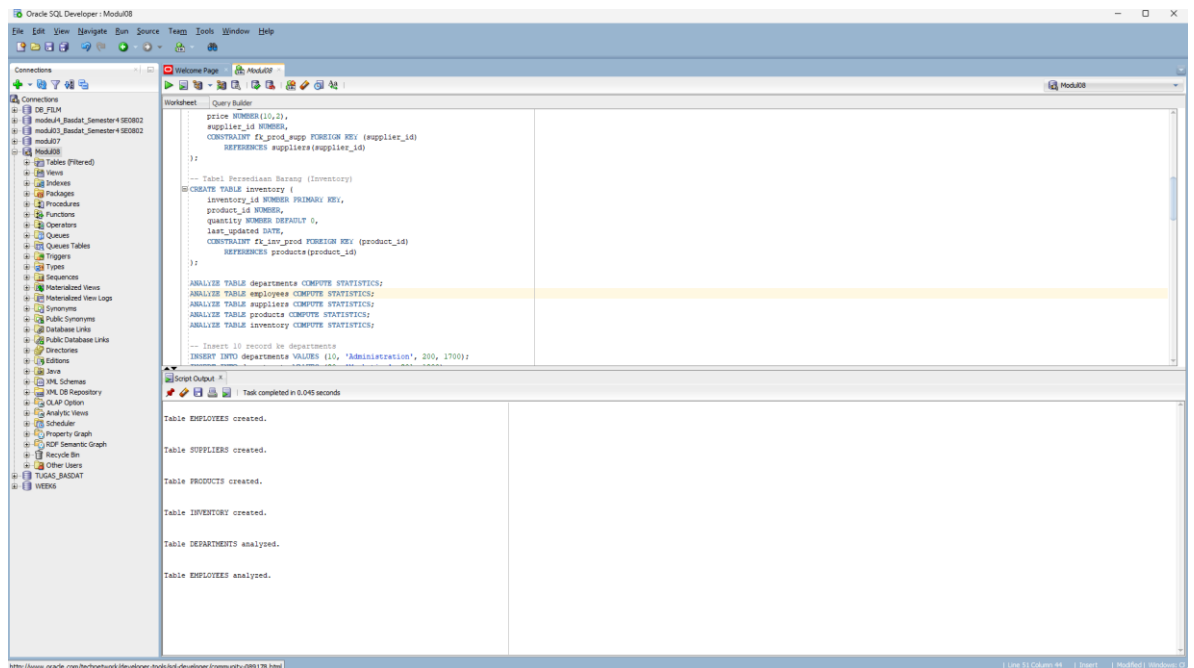


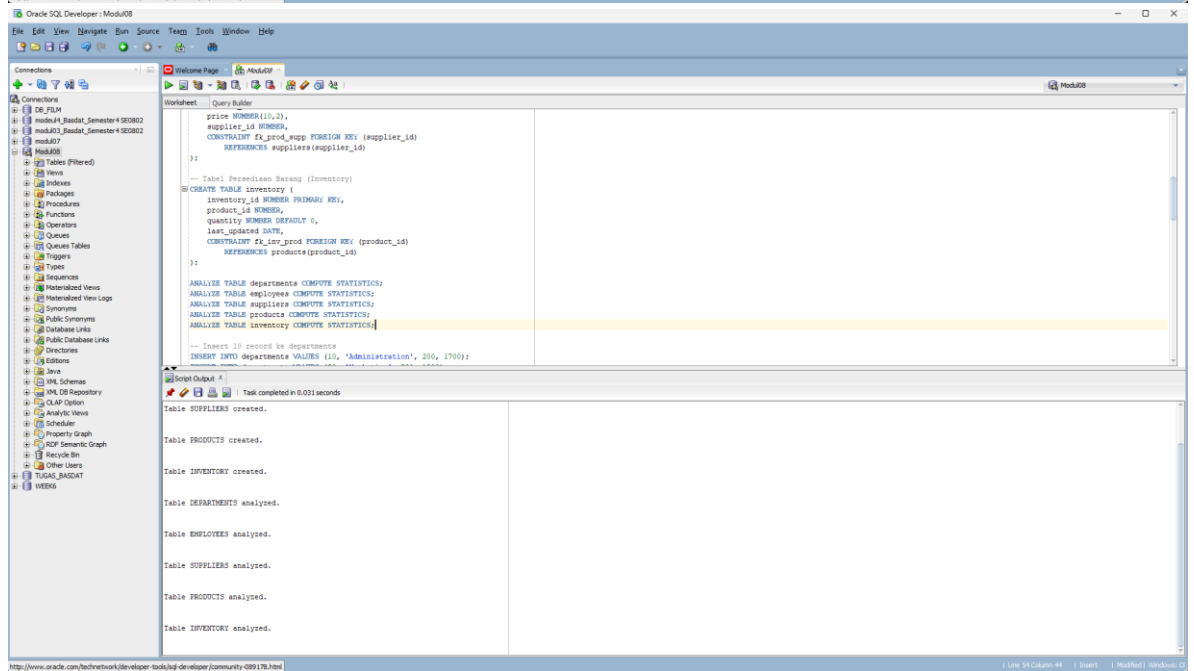
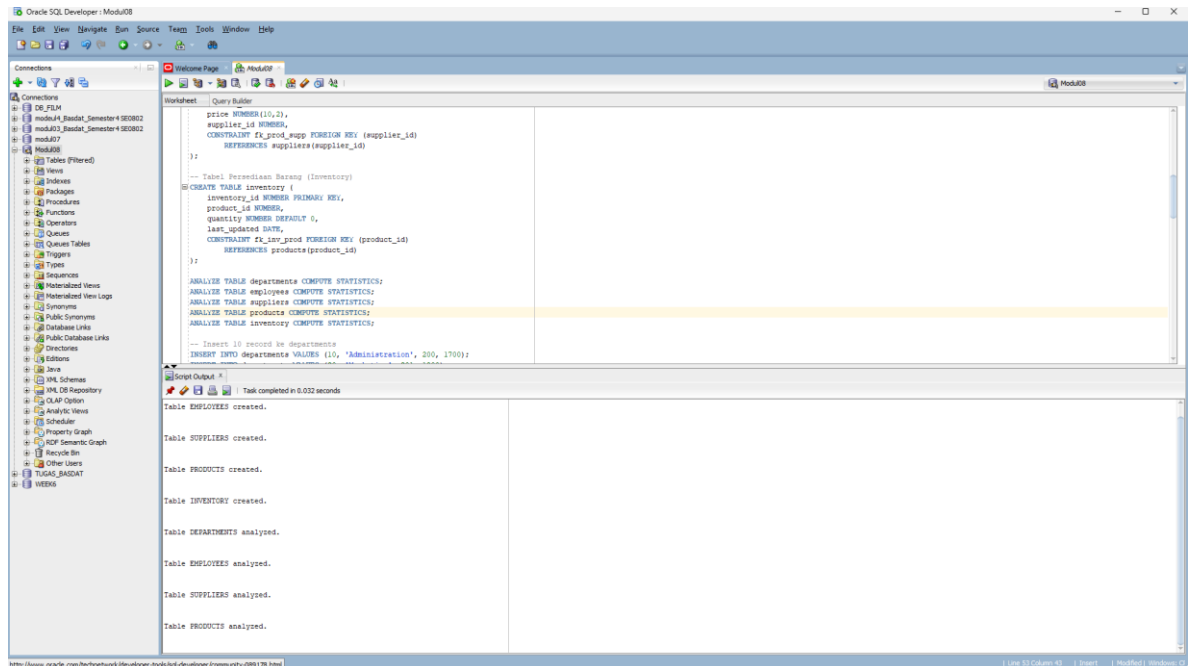


2. Analisis table-table dengan compute statistics

Jawaban:

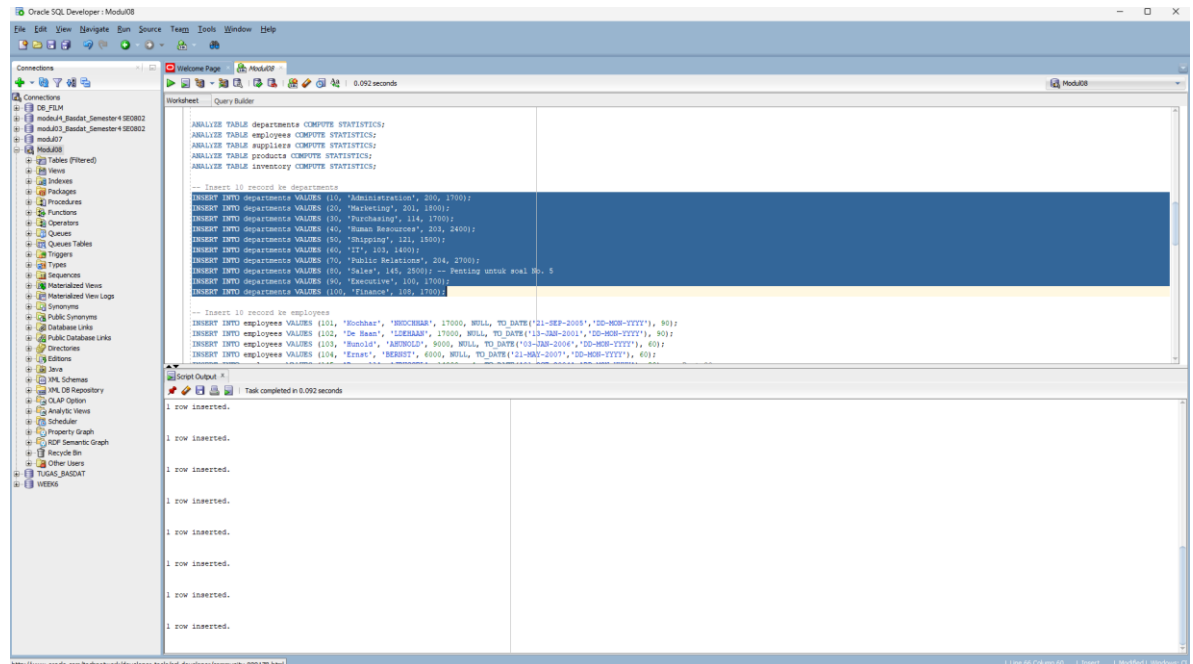






3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

Jawaban:



The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' window displays a SQL script with the following content:

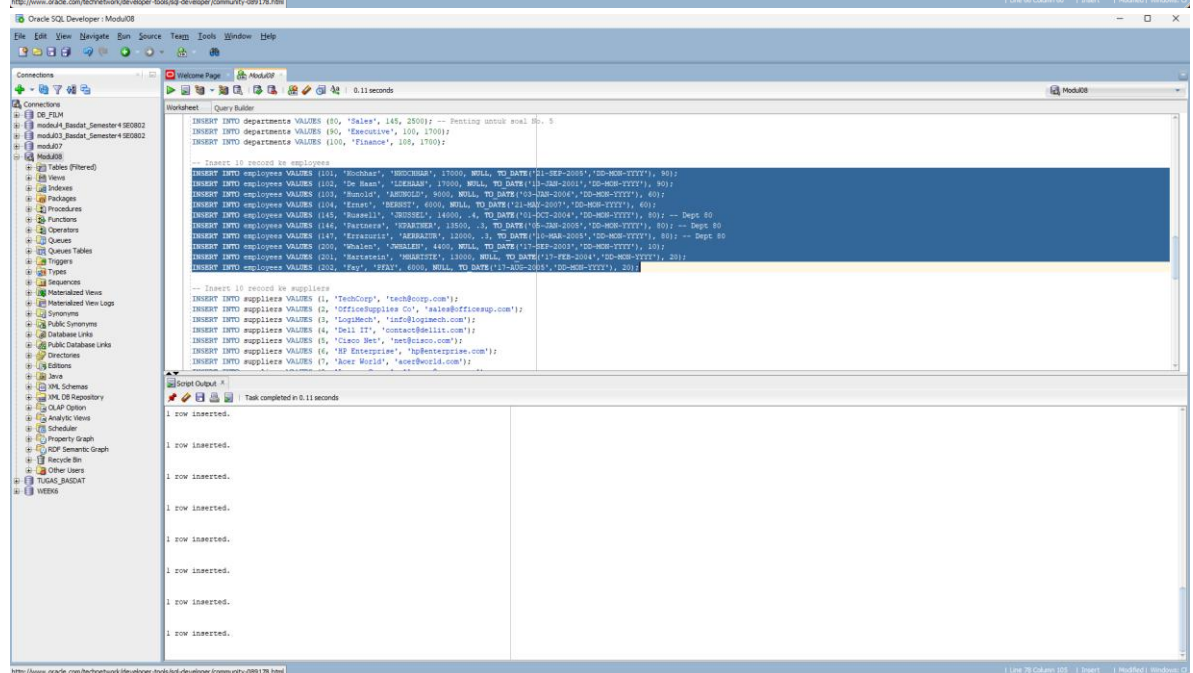
```
ANALYZE TABLE departments COMPUTE STATISTICS;
ANALYZE TABLE employees COMPUTE STATISTICS;
ANALYZE TABLE suppliers COMPUTE STATISTICS;
ANALYZE TABLE products COMPUTE STATISTICS;
ANALYZE TABLE inventory COMPUTE STATISTICS;

-- Insert 10 record ke departments
INSERT INTO departments VALUES (10, 'Administration', 800, 2700);
INSERT INTO departments VALUES (20, 'Marketing', 201, 1800);
INSERT INTO departments VALUES (30, 'Purchasing', 114, 1700);
INSERT INTO departments VALUES (40, 'Human Resources', 201, 2400);
INSERT INTO departments VALUES (50, 'Shipping', 121, 1800);
INSERT INTO departments VALUES (60, 'IT', 103, 1400);
INSERT INTO departments VALUES (70, 'Public Relations', 204, 2700);
INSERT INTO departments VALUES (80, 'Sales', 145, 2500); -- Pending untuk soal No. 5
INSERT INTO departments VALUES (90, 'Executive', 106, 1700);
INSERT INTO departments VALUES (100, 'Finance', 108, 1700);

-- Insert 10 record ke employees
INSERT INTO employees VALUES (101, 'Hochberg', 'HUCHBERG', 17000, NULL, TO_DATE('21-SEP-2005', 'DD-MON-YYYY'), 80);
INSERT INTO employees VALUES (102, 'De Haan', 'DEHAAN', 17000, NULL, TO_DATE('11-JAN-2001', 'DD-MON-YYYY'), 90);
INSERT INTO employees VALUES (103, 'Rudolfs', 'RUDOLFS', 9000, NULL, TO_DATE('13-JAN-2004', 'DD-MON-YYYY'), 90);
INSERT INTO employees VALUES (104, 'Fey', 'FEY', 6000, NULL, TO_DATE('12-MAY-2007', 'DD-MON-YYYY'), 60);
INSERT INTO employees VALUES (105, 'Fuchs', 'FUCHS', 4000, NULL, TO_DATE('12-MAY-2007', 'DD-MON-YYYY'), 60);
INSERT INTO employees VALUES (106, 'Mavris', 'MAVRIS', 14000, 1, TO_DATE('01-OCT-2004', 'DD-MON-YYYY'), 80); -- Dept 80
INSERT INTO employees VALUES (107, 'Patel', 'PATEL', 15000, 1, TO_DATE('05-MAR-2005', 'DD-MON-YYYY'), 80); -- Dept 80
INSERT INTO employees VALUES (108, 'Frazar', 'FRAZAR', 12000, 1, TO_DATE('10-MAR-2004', 'DD-MON-YYYY'), 80); -- Dept 80
INSERT INTO employees VALUES (109, 'Whalen', 'WHALEN', 4000, NULL, TO_DATE('17-SEP-2003', 'DD-MON-YYYY'), 10);
INSERT INTO employees VALUES (110, 'Marsden', 'MARSDEN', 18000, NULL, TO_DATE('11-FEB-2004', 'DD-MON-YYYY'), 20);
INSERT INTO employees VALUES (111, 'Pyey', 'PYEY', 6000, NULL, TO_DATE('11-JUN-2004', 'DD-MON-YYYY'), 20);

-- Insert 10 record ke suppliers
INSERT INTO suppliers VALUES (1, 'TechCorp', 'tech@corp.com');
INSERT INTO suppliers VALUES (2, 'OfficeSupplies Co', 'sales@officesup.com');
INSERT INTO suppliers VALUES (3, 'LogiTech', 'info@logitech.com');
INSERT INTO suppliers VALUES (4, 'Mail IT', 'contact@ballit.com');
INSERT INTO suppliers VALUES (5, 'Cisco Int', 'net@cisco.com');
INSERT INTO suppliers VALUES (6, 'ID Enterprise', 'help@enterprise.com');
INSERT INTO suppliers VALUES (7, 'Ace World', 'aceworld.com');
```

The 'Script Output' window shows the results of the execution, indicating that 10 rows were inserted into each of the three tables (departments, employees, and suppliers).



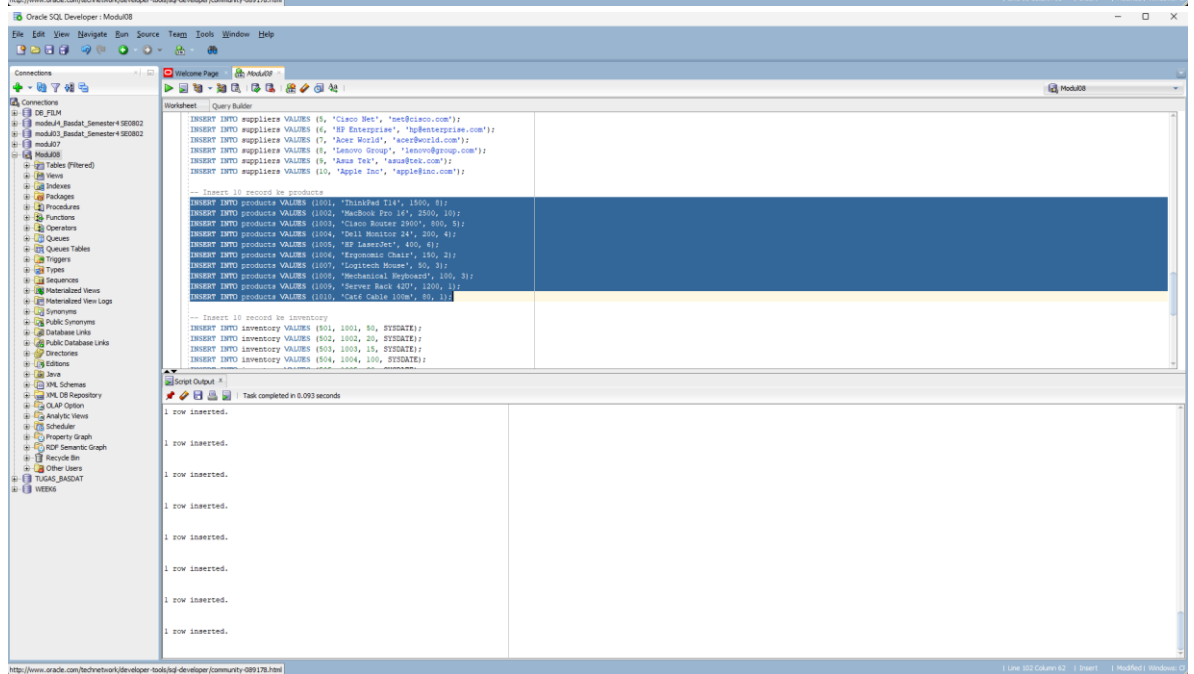
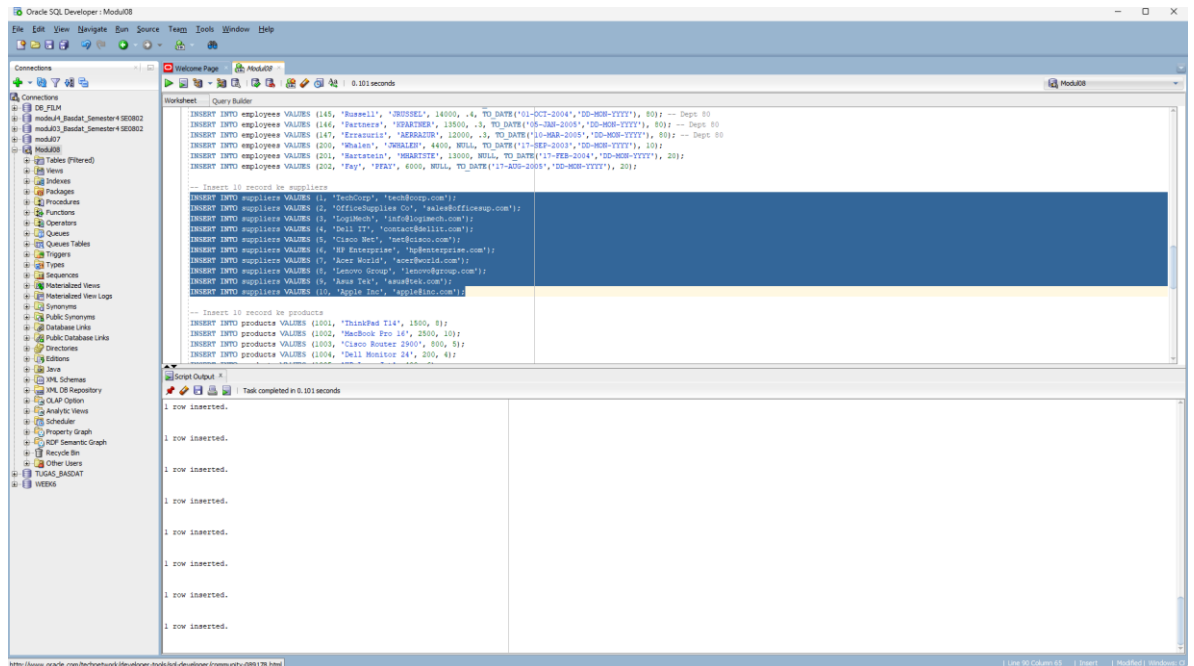
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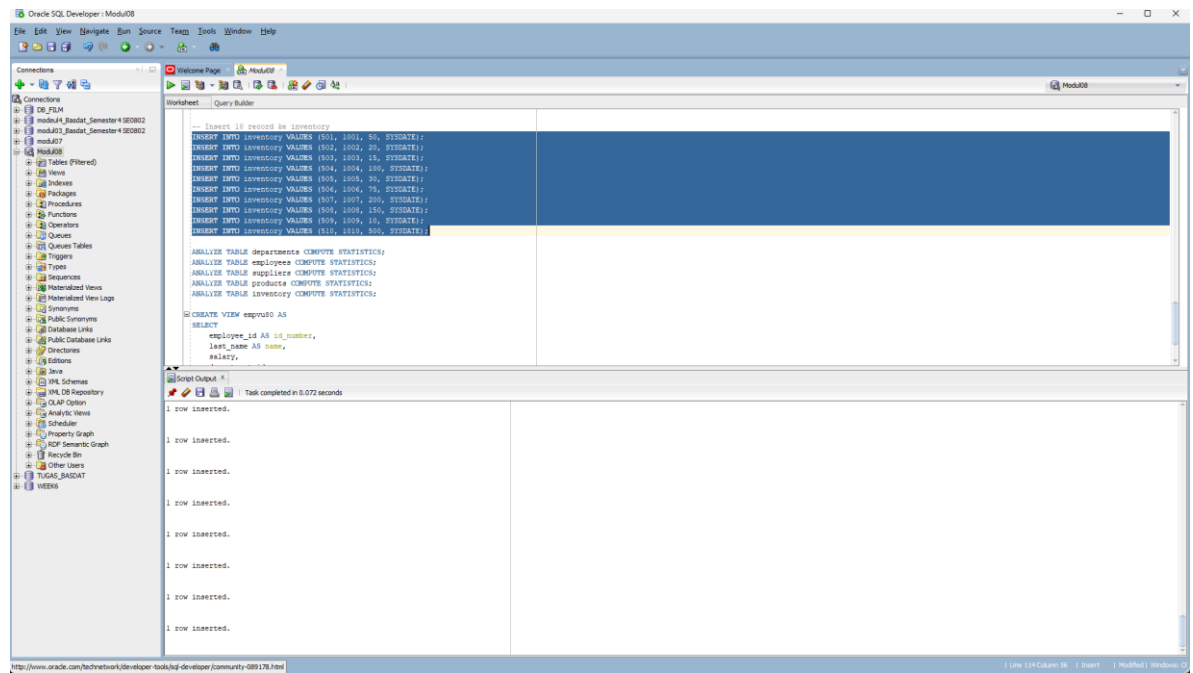
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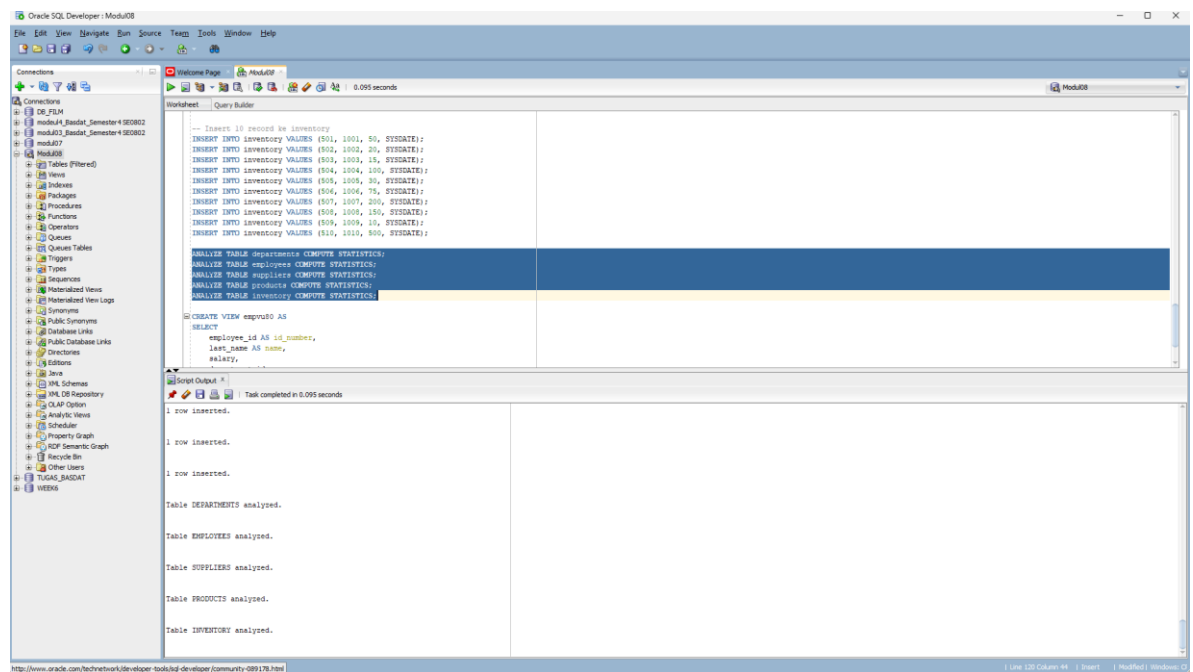
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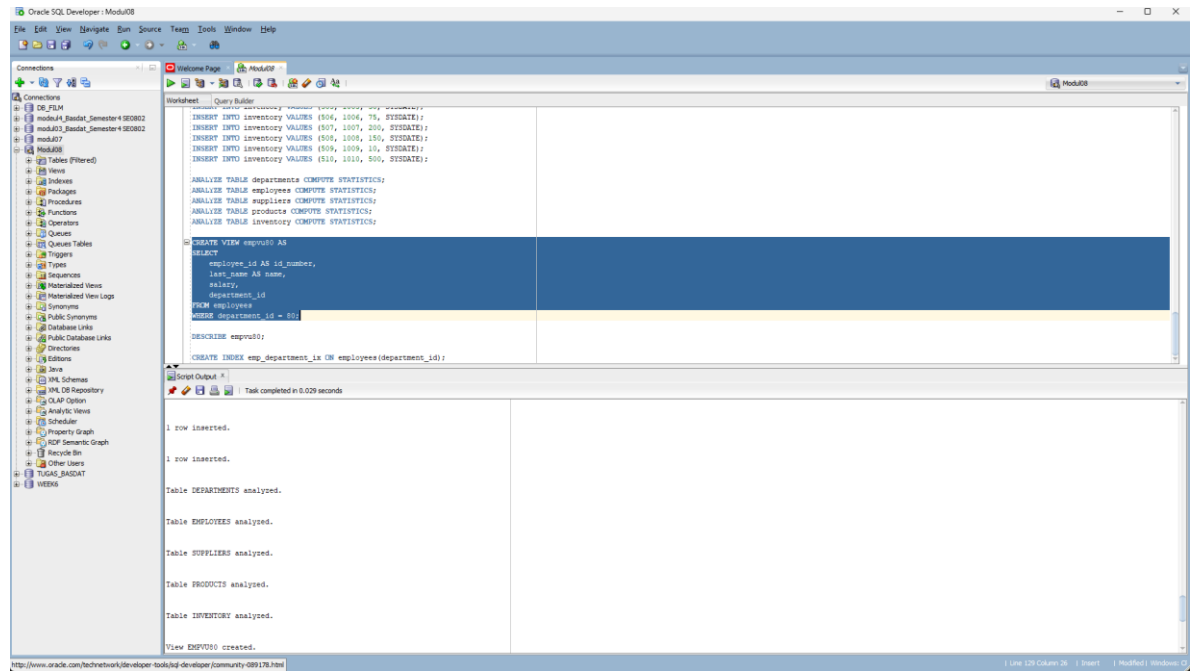
4. Analisis table-table dengan compute statistics

Jawaban:



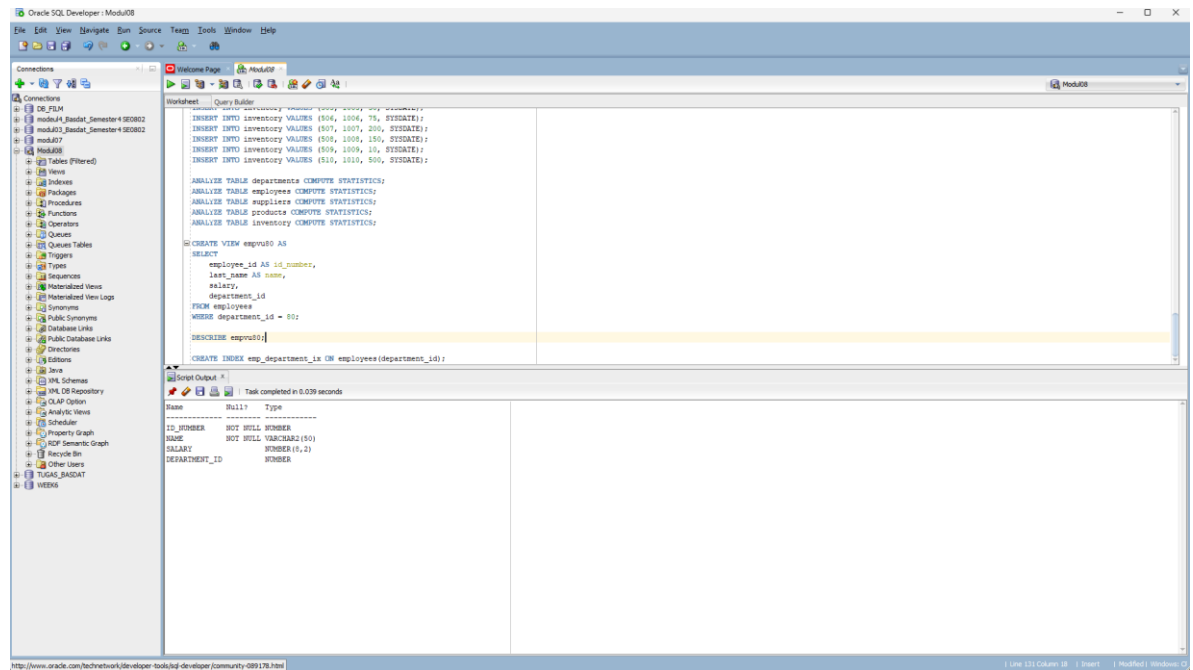
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

Jawaban:



6. Kemudian tampilkan struktur dari view empvu80.

Jawaban:



7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Jawaban:

